

Problem Set 1

Exercise 1 Describe a hypothetical ideal randomized controlled experiment to study the effect of the consumption of alcohol on long-term memory loss. Suggest some impediments to implementing this experiment in practice.

Exercise 2 Show the identities:

$$(a) \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^n (y_i - \bar{y})x_i$$

$$(b) \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \bar{x})x_i$$

Exercise 3 Consider two random variables, X and Y . Suppose that Y takes on k values $y_1, \dots, y_k$ and that X takes on l values $x_1, \dots, x_l$.

- (a) Show that $Pr(Y = y_i) = \sum_{j=1}^l Pr(Y = y_j | X = x_i)Pr(X = x_i)$
- (b) Verify that $\mathbb{E}(Y) = \sum_{i=1}^l \mathbb{E}(Y | X = x_i)Pr(X = x_i)$
- (c) Suppose that X and Y are independent. Show that $\sigma_{XY} = 0$ and $\rho_{XY} = 0$

Exercise 4 The following table gives the joint probability distribution between employment status and college graduation among those either employed or looking for work (unemployed) in the working-age population of South Africa.

	Unemployed ($Y = 0$)	Employed ($Y = 1$)	Total
Non-College Graduates ($X = 0$)	0.078	0.673	0.751
College Graduates ($X = 1$)	0.042	0.207	0.249
Total	0.12	0.88	1

- (a) Compute $\mathbb{E}(Y)$.
- (b) The unemployment rate is the fraction of the labor force that is unemployed. Show that the unemployment rate is given by $1 - \mathbb{E}(Y)$.
- (c) Calculate $\mathbb{E}(Y | X = 1)$ and $\mathbb{E}(Y | X = 0)$.
- (d) Calculate the unemployment rate for (i) college graduates and (ii) non-college graduates.
- (e) A randomly selected member of this population reports being unemployed. What is the probability that this worker is a college graduate? A non-college graduate?
- (f) Are educational achievement and employment status independent? Explain.

Exercise 5 Consider the problem of predicting Y using another variable, X , so that the prediction of Y is some function of X , say $g(X)$. Suppose that the quality of the prediction is measured by the squared prediction error made on average over all predictions, that is, by $\mathbb{E}\{[Y - g(X)]^2\}$. This exercise provides a non-calculus proof that of all possible prediction functions g , the best prediction is made by the conditional expectation, $\mathbb{E}(Y | X)$.

- (a) Let $\hat{Y} = \mathbb{E}(Y | X)$ and $u = Y - \hat{Y}$ denote its prediction error. Show that $\mathbb{E}(u) = 0$.
- (b) Show that $\mathbb{E}(uX) = 0$.
- (c) Let $\tilde{Y} = g(x)$ denote a different prediction of Y using X , and let $v = Y - \tilde{Y}$. Show that $\mathbb{E}[(Y - \tilde{Y})^2] > \mathbb{E}[(Y - \hat{Y})^2]$

Exercise 6 Let Y be a Bernoulli random variable with success probability $\Pr(Y = 1) = p$, and let $Y_1, \dots, Y_n$ be i.i.d. draws from this distribution. Let $\hat{p}$ be the fraction of successes (1s) in this sample.

- (a) Show that $\hat{p} = \bar{Y}$.
- (b) Show that $\hat{p}$ is an unbiased estimator of p .
- (c) Show that $\text{var}(\hat{p}) = p(1 - p)/n$

Exercise 7 Consider the estimator $\tilde{Y} = \frac{1}{n}(\frac{1}{2}Y_1 + \frac{3}{n}Y_2 + \frac{1}{2}Y_3 + \frac{3}{n}Y_4 + \dots + \frac{1}{2}Y_{n-1} + \frac{3}{n}Y_n)$.

- (a) Show that $\mathbb{E}(\tilde{Y}) = \mu_Y$
- (b) Show that $\text{var}(\tilde{Y}) = 1.25\sigma_Y^2/n$

Exercise 8 This exercise shows that the sample variance is an unbiased estimator of the population variance when $Y_1, \dots, Y_n$ are i.i.d. with mean μ_Y and variance σ_Y^2 .

- (a) Using the following identity - $\text{var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$; show that $\mathbb{E}(Y_i - \bar{Y})^2 = \text{var}(Y_i) + \text{var}(\bar{Y}) - 2\text{cov}(Y_i, \bar{Y})$
- (b) Using the following identity - $\text{cov}(a + bX + cY, Y) = b\sigma_{XY} + c\sigma_Y^2$; show that $\text{cov}(Y_i, \bar{Y}) = \sigma_Y^2/n$
- (c) Use your previous answers to show that $\mathbb{E}(s_Y^2) = \sigma_Y^2$

Exercise 9 Suppose (X_i, Y_i) are i.i.d. with finite fourth moments. Prove that the sample covariance is a consistent estimator of the population covariance.

Exercise 10 Suppose Y_i are i.i.d. and $Y_i \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ for $i = 1, \dots, n$. With σ_Y^2 known, the t -statistic for testing $H_0 =: \mu_Y = 0$ vs. $H_1 =: \mu_Y > 0$ is $t = \bar{Y}/SE(\bar{Y})$ with $SE(\bar{Y}) = \sigma_Y/\sqrt{n}$. Suppose that $\sigma_Y = 10$ and $n = 100$ so that $SE(\bar{Y}) = 1$. Using a test with a size of 5%, the null hypothesis is rejected if $|t| > 1.64$ (one-sided test).

- (a) Suppose $\mu_Y = 0$, so the null hypothesis is true. What is the probability that the null hypothesis is rejected?
- (b) Suppose $\mu_Y = 2$, so the alternative hypothesis is true. What is the probability that the null hypothesis is rejected?
- (c) Suppose that in 90% of cases the data are drawn from a population where the null is true ($\mu_Y = 0$) and in 10% of cases the data come from a population where the alternative is true and $\mu_Y = 2$. Your data came from either the first or the second population, but you don't know which.

(i) You compute the t -statistic. What is the probability that $|t| > 1.64$ – that is, that you reject the null hypothesis?

(ii) Suppose you reject the null hypothesis; that is, $|t| > 1.64$. What is the probability that the sample data were drawn from the $\mu_Y = 0$ population?

(d) It is hard to discover a new effective drug. Suppose 90% of new drugs are ineffective and only 10% are effective. Let Y denote the drop in the level of a specific blood toxin for a patient taking a new drug. If the drug is ineffective, $\mu_Y = 0$ and $\sigma_Y = 10$; if the drug is effective, $\mu_Y = 2$ and $\sigma_Y = 10$.

(i) A new drug is tested on a random sample of $n = 100$ patients, data are collected, and the resulting t -statistic is found to be greater than 1.64. What is the probability that the drug is ineffective (i.e., what is the false positive rate for the test using $|t| > 1.64$)?

(ii) Suppose the one-sided test uses instead the 0.5% significance level. What is the probability that the drug is ineffective (i.e., what is the false positive rate)?